quantmsdiann: a scalable SDRF-driven DIA-NN workflow for reanalysis of single-cell, spatial, and bulk proteomics datasets

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Abstract

Public proteomics archives now host thousands of data-independent acquisition (DIA) datasets, but they remain difficult to integrate and reuse because each dataset has been processed with a different software configuration, and most of them lack standardised metadata. Here, we present `quantmsdiann`, an open-source Nextflow/nf-core workflow that parallelises `DIA-NN` across cloud and high-performance computing (HPC) infrastructures, guided by the experimental design declared in SDRF format. The workflow ships container-pinned profiles for each supported `DIA-NN` version through Docker and Singularity, with no manual installation or dependency resolution, and natively handles all vendor formats and the HUPO-PSI standard mzML. It exports harmonised quantification tables as MSstats input, the new Quantitative Proteomics eXchange format (QPX), and a `pmultiqc` quality-control report. The parallelisation design reanalyses a 2,300-run single-cell dataset in 2.2 hours on an HPC cluster of 300 nodes. We benchmark `quantmsdiann` on four ProteoBench DIA instrument modalities, where distributed `quantmsdiann` reproduces single-machine community submissions independent of `DIA-NN` release. We reanalyse public DIA deposits spanning single-cell proteomes (HeLa on Orbitrap Astral, PXD046357; and an oocyte plexDIA cohort, MSV000093870) and bulk cell-line panels (NCI-60/PXD003539 and ProCan/PXD030304, with three further cohorts

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combined into a five-cohort pan-cohort under a single invocation). Reprocessing with current DIA-NN releases matches or exceeds the originally deposited counts: across the bulk cell-line and HeLa single-cell cohorts it recovers more protein groups than the original or 1.8.1 analyses (gains of +17% to +59% across cohorts, each under the original study's own counting criterion), as well as more precursors wherever a DIA-NN-based original count is available. In addition, we demonstrate the scalability and support of quantmsdiann for plexDIA and phosphoproteomics experiments. quantmsdiann is a step towards scalable, reproducible reanalysis of the growing DIA data landscape and a foundation for community efforts in large-scale DIA meta-analysis and atlas building. quantmsdiann is available at https://github.com/bigbio/quantmsdiann.

Keywords: DIA-NN, single-cell proteomics, spatial proteomics, phosphoproteomics, plexDIA, data-independent acquisition, Nextflow, nf-core, SDRF, reproducibility, QPX, pmultiqc

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1. Introduction

Data-independent acquisition (DIA) is a rapidly growing mass-spectrometry strategy for proteomics at scale, now approaching parity with data-dependent acquisition among public proteomics deposits, and DIA-NN [1] is among the most widely adopted analysis engines across timsTOF [2, 3], Astral, and Orbitrap platforms. Recent benchmarks confirm DIA-NN's competitiveness for single-cell DIA workflows [4], and recent releases have expanded support for plexDIA, timsTOF parallel accumulation-serial fragmentation (PASEF), and vendor-native raw-file reading. At the same time, DIA dataset sizes and run counts are growing rapidly [5], and the proteomics community is increasingly interested in reanalysing public deposits to recover additional identifications, integrate across cohorts, and build large-scale DIA atlases [6, 7].

DIA-NN's command-line interface, simplicity, and analytical power make it a natural engine for mining public archives, but realising it at scale is fundamentally a distributed-computing problem: extracting more information from public data means reprocessing cohorts of hundreds to thousands of runs fast enough that reanalysis can be iterative rather than a one-off effort. The same need extends beyond reanalysis: laboratories generating their own large cohorts increasingly have access to cloud and high-performance computing (HPC) infrastructure and benefit from distributing the analysis rather than running it on a single workstation. The steps of a DIA-NN run are heterogeneous and map naturally onto heterogeneous hardware: many lightweight per-file passes that fit small nodes, and a few memory- and compute-heavy cohort-level steps that need large nodes. Efficient large-scale analysis therefore depends on software containers and workflow engines that can place each task on appropriate cloud or HPC resources [8]. Despite this analytical maturity, most DIA-NN users run the tool as a desktop application, which limits its scalability for large datasets. Deployments on HPC and cloud infrastructure remain difficult, relying on shell scripts, bespoke directory conventions, and hand-curated sample sheets. As a result, the same raw data analysed by two laboratories, or with two DIA-NN releases, can yield non-overlapping quantification tables that are difficult to reconcile.

In 2024 we published bigbio/quantms [6], a Nextflow pipeline that standardised multi-engine DIA and DDA analyses behind a Sample and Data Relationship Format (SDRF) [9] input on the nf-core framework [10]. quantms was, however, built around DIA-NN 1.8.1 with a fixed parameter set, and its DDA/DIA combination within the same workflow made extending DIA-specific functionality difficult. New DIA-NN parameters conflicted with DDA options, versions were hard-coded, sample-level acquisition parameters were not properly propagated to per-run DIA-NN invocations, and raw-file conversion was mandatory. Iterative reanalysis at archive scale requires a workflow that combines the reproducibility of nf-core, the sensitivity of current DIA-NN releases, the metadata discipline of SDRF, and a parallelisation strategy capable of processing thousands of raw files.

Here, we present quantmsdiann, an independent SDRF-driven Nextflow pipeline dedicated to DIA-NN that extends DIA-specific features beyond what the original bigbio/quantms pipeline could accommodate. While quantmsdiann shares its SDRF-first design philosophy and parts of the configuration-generation tooling with quantms, its workflow, container set, and module library are developed and released independently, with container-pinned profiles for every DIA-NN version selectable at runtime under Docker or Singularity. Downstream, it exports a harmonised QPX (Quantitative Proteomics eXchange) Parquet archive alongside the standard DIA-NN reports and MSstats input. We demonstrate the pipeline along three axes: ProteoBench [11] benchmarks across four DIA modalities and successive DIA-NN releases; independent reanalyses of public DIA deposits spanning bulk cell lines (NCI-60, ProCan), single-cell datasets, a spatial Deep Visual Proteomics cohort, and a phosphoproteomics cohort; and a pan-cohort of five public DIA cell-line reanalyses integrated under a single SDRF-driven invocation. Across the single-cell and bulk cell-line cohorts, reanalysis with the current DIA-NN build recovers on average ∼36% more protein groups (+17% to +59% across cohorts) than the original or 1.8.1 analyses, so reanalysing archived data with an up-to-date engine recovers measurably more of each deposit.

2. Experimental Procedures

2.1. Pipeline implementation

quantmsdiann is implemented in Nextflow DSL2 (≥25.10.4) and follows nf-core community standards [10] for pipeline structure, testing, documentation, and continuous integration. It supports both DIA-NN library-free analysis (predicting the spectral library in-silico from a FASTA reference) and library-based analysis against a user-supplied spectral library; all analyses in this study used the library-free mode; the reanalysis cohorts used the UniProt human reference proteome [12] (UP000005640, downloaded March 2026), while the ProteoBench HYE benchmarks used the corresponding three-species (human, yeast, *E. coli*) FASTA. Each computational step is packaged as a versioned container image (Docker, Singularity, or Apptainer). Open-source community tools (ThermoRawFileParser [13], pridepy [14], QPX, and the wiffconverter SCIEX reader) are pulled from public registries (BioContainers [15] or ghcr.io/bigbio). Because the DIA-NN license restricts redistribution, only version 1.8.1 is shipped as an image (the default, from BioContainers); releases from 1.9 onward (1.9.2–2.5.1, including the 2.5.1-enterprise build) are built locally from the per-release recipes in the companion quantms-containers repository (https://github.com/bigbio/quantms-containers) and tagged so the workflow selects them automatically (Supplementary Note 1). The repository at https://github.com/bigbio/quantmsdiann is organised as a standard nf-core pipeline with main workflow (main.nf), subworkflows, modules, test data and CI configurations, and documentation. Pipeline version v2.2.0 (release codename "Chongqing", 2026-06-16) was used for the analyses reported in this paper.

2.2. SDRF extensions for quantmsdiann

quantmsdiann accepts SDRF [9] as primary input. SDRF parsing is performed using the sdrf-pipelines Python package developed within the quantms project, which validates the SDRF against the proteomics-specific subset of EFO, PSI-MS, and PRIDE ontology terms and emits a normalised per-sample channel for downstream Nextflow processes. To express DIA- and DIA-NN-specific acquisition settings at the sample level, we extended the SDRF with additional comment columns: the MS1 and MS2 scan mass-window intervals and, for multiplexed experiments, the plexDIA channel definitions and their isotopologue labels. Because these settings are declared per sample, heterogeneous MS1/MS2 window schemes, or a plexDIA channel layout, can be mixed within a single cohort without any change to the workflow. We added a `convert-diann` command to sdrf-pipelines that reads these extended columns together with the standard fields (enzyme, fixed and variable modifications, instrument, precursor mass tolerance, fragment mass tolerance, and missed-cleavage settings) and writes a per-run `DIA-NN` command-line configuration; where SDRF columns are absent, `DIA-NN` defaults appropriate for the inferred instrument are applied.

2.3. Raw file handling and `DIA-NN` *execution*

A key advance over its predecessor quantms is that native vendor (raw) formats are first-class inputs in `quantmsdiann`: whereas quantms converted every raw file to mzML up front, `quantmsdiann` reads each instrument's native format directly and converts only when a particular format or `DIA-NN` version requires it. Thermo `.raw` or timsTOF (.d or .d.zip) files are read natively by `DIA-NN` $\geq$ 2.1.0; for DIA-NN versions lower than 2.1.0 conversion to mzML is performed with ThermoRawFileParser [13] (`-mzml_convert`); Bruker `.d` folders are ingested natively, with compressed forms (`.d.tar`/`.zip`/`.gz`) decompressed first; and Sciex `.wiff` / `.wiff2` are converted to mzML with the open-source `wiffconverter` (`https://github.com/bigbio/wiffconverter`). Generic `.dia` and pre-converted `.mzML` inputs pass through unchanged, with optional OpenMS re-indexing (`-reindex_mzml`).

2.4. Quality control and QPX *export*

Per-run and per-cohort quality control reports are generated by `pmultiqc` [16], a proteomics extension of MultiQC [17] that summarises `DIA-NN`-specific metrics including peptide and protein counts, missed cleavages, charge-state distributions, precursor and fragment mass accuracy, and per-run identification yield. The QPX archive is produced by the `qpxc` CLI from the `qpx` package (`https://github.com/bigbio/qpx`), which writes Parquet files for identifications (psm, peptide, protein-group, feature), metadata (sample, run, ontology), and provenance, and optionally exports MuData (`.h5mu`) containers for scverse-based [18] downstream analysis. The aim of exporting to QPX is to bundle the SDRF, identifications, quantifications, and pipeline provenance into a single queryable artefact for reanalysis and long-term archival, and to provide a standardised input for downstream tools that support the format (e.g. MSstats, Perseus, or custom scripts).

2.5. Datasets benchmarked and compute environment

Supplementary Note 2 (Table S1) lists all benchmark and reanalysis datasets with their instrument, acquisition mode, and deposition year, and an SDRF prepared from the original sample metadata. They fall into four experiment classes: *ProteoBench* DIA benchmarks (modules 5, 7, 9, and 10); *single-cell and spatial proteomics*, comprising the HeLa single-cell cohort PXD046357 (Orbitrap Astral), the oocyte plexDIA cohort MSV000093870, the PXD064049 spatial Deep Visual Proteomics cohort, and the PXD071075 single-cell scalability sweep; *bulk cell-line panels* (PXD003539, PXD030304, PXD004701, PXD041421, and PXD017199); and *phosphoproteomics* (PXD034128, PXD034623, and PXD049692). For each cell-line reanalysis cohort, the original published quantification was downloaded from a public repository as the comparator. `quantmsdiann` counts follow a single global rule (protein groups at `Lib.PG.Q.Value` ≤ 0.01, precursors at `Lib.Q.Value` ≤ 0.01, decoys dropped, no contaminant or positive-quantity filter); the like-for-like-threshold rule applied to each original headline count is described in Supplementary Note 3. Runtime benchmarks were run on the EMBL-EBI HPC cluster (LSF scheduler), with per-task CPU and memory dispatched by Nextflow and memory profiled from the Nextflow trace report (`-with-trace`).

3. Results

3.1. `quantmsdiann` pipeline architecture

Built on the nf-core framework [10], `quantmsdiann` accepts a single SDRF file as primary input, which encodes per-sample metadata (cell line or tissue origin, factor values, instrument, post-translational modifications) and per-run file references. We extended the SDRF with DIA-relevant columns (MS1 and MS2 mass windows, and plexDIA channel definitions and labels) and added an adapter to sdrf-pipelines [9] (`convert-diann`) that translates each run's declared metadata into a per-run `DIA-NN` command-line configuration. Every run is therefore parametrised from its own metadata with no manual configuration or parameter-harmonisation step, removing a common source of analysis-script divergence between groups: different MS1/MS2 window schemes can be mixed within a cohort, or a plexDIA dataset run with its channel definitions, without any workflow change beyond the SDRF. The workflow comprises four mandatory and three optional modules organised around the `DIA-NN` analysis core (Fig. 1), alternating between per-file parallel stages (red; raw-file preparation, preliminary calibration, and individual search) and collective stages (green; in-silico library generation, empirical library assembly, final quantification, MSstats conversion, and `pmultiqc` [16] quality-control reporting) operating on the full cohort.

`quantmsdiann` follows `DIA-NN`'s distributed multi-pass flow, alternating parallel per-file work with serial cohort-wide steps (Fig. 1). Each raw file is first prepared *in parallel* (vendor-format conversion, decompression, or indexing) where needed; conversion to mzML is mandatory for early `DIA-NN` versions and for some vendor formats such as SCIEX `.wiff`. A spectral library is then predicted in-silico from the FASTA once (*serial*; skipped when a library is supplied), and every file is calibrated against it in a *parallel* first pass. The first-pass results are merged into a single empirical library (*serial*); each file is searched again against that library *in parallel*; and a final consolidation pass merges all per-file results into the cohort quantification, applying match-between-runs, normalisation, and protein inference (*serial*). MSstats conversion, `pmultiqc` reporting, and QPX export then run once at the end (*serial*). The two per-file search passes carry essentially all of the compute and scale across cluster nodes, driving the throughput reported below (Section 3.2); the serial steps set a largely fixed wall-clock floor. Outputs include the standard `DIA-NN` report tables: `report.tsv` (or `report.parquet`), `report.pr_matrix.tsv`, and `report.pg_matrix.tsv`, alongside MSstats-formatted input and a `pmultiqc` [16] quality-control report. A harmonised QPX archive bundles identifications, metadata, and provenance into a Parquet-based artefact, queried directly with DuckDB and, for the quantification layers, exported to MuData (`.h5mu`) for scverse-compatible [18] downstream analysis.

We evaluate `quantmsdiann` along four lines that together motivate it as a reanalysis platform. First, because every release is provided as a container-pinned profile selectable at runtime, the same SDRF can be analysed with any supported `DIA-NN` version (1.8.1 through 2.5.1, including the enterprise build, and any other release via a custom container), so the engine version is a runtime choice rather than a reinstallation. Second, the per-file parallel design lets a single invocation scale with data volume, from six-run benchmarks to cohorts of thousands of single-cell runs (Section 3.2). Third, on the ProteoBench community benchmarks this distributed execution reproduces the identification and quantification performance of single-machine `DIA-NN` runs, i.e. scaling out does not change the result relative to a desktop analysis (Section 3.2). Fourth, with that equivalence established, reanalysing public deposits with current `DIA-NN` releases recovers more identifications than the originally deposited analyses, with the largest gains on low-input single-cell data (Sections 3.3–3.4).

3.2. quantmsdiann throughput, scaling, and analytical equivalence

To characterise how `quantmsdiann` scales on an HPC cluster or cloud, we simulated the reanalysis of an entire single-cell experiment of ~2,300 raw files (PXD071075) across cluster sizes from 10 to 300 nodes (set through the Nextflow `executor.queueSize`), and measured the end-to-end wall-clock time (Fig. 2a). Because the per-file work is parallelised across nodes, a single SDRF-driven invocation completed the whole cohort within hours: wall-clock fell monotonically from 37.4 h at 10 nodes to 2.2 h at 300 nodes (excluding the one-time in-silico library-prediction step, which is independent of cluster size). Against an ideal linear ($1/N$) strong-scaling reference the measured curve stays close up to ~100 nodes (parallel efficiency 95% at 50 nodes, 82% at 100) and then bends away, with a practical knee near 200 nodes (2.4 h, 79% efficiency): beyond it a 30× increase in cluster width yields only a 17× speed-up, and efficiency falls to 58% at 300 nodes. This floor reflects two effects a single sweep cannot separate: the serial cohort-level steps that run once regardless of node count (empirical-library merge, final consolidation, export), and contention for filesystem I/O and scheduler slots as concurrency rises on a shared cluster, so part of the plateau is specific to this EMBL-EBI allocation rather than an intrinsic limit of the workflow. With one run per sweep point the exact knee is only indicative, but the practical message holds: size the cluster to the desired turn-around time rather than to maximum throughput.

We then compared the time-to-finish across every dataset in this study (Fig. 2b), reporting each run net of the one-time in-silico library-prediction step. The in-silico library generation is excluded because it cannot be parallelised and depends heavily on the library-generation parameters (e.g. the number of modifications and the MS1 and MS2 window scheme); the sole exception is PXD003539, whose incomplete execution trace did not allow the library step to be separated, so its bar conservatively includes it (a ∼0.4 h overstatement relative to the other bars). Across cohorts spanning three orders of magnitude in file count, wall-clock stayed within hours because the per-file passes absorb additional files in parallel: the 5,798-run ProCan cohort (PXD030304) finished in 6.2 h, in the same band as cohorts orders of magnitude smaller, while the 6-run ProteoBench benchmarks completed in under 0.7 h. Time-to-finish is therefore governed by per-run search cost and cohort-level consolidation rather than by the number of files: the longest reanalysis was a 206-run cohort on an older Q Exactive instrument (PXD017199, 11.4 h), exceeding the far larger ProCan cohort, followed by a 10-file phosphoproteomics cohort whose deep variable-modification search drives its 8.0 h (PXD049692). Peak memory during the cohort-level consolidation stayed within the 64–128 GB available on commodity HPC nodes, so no high-memory specialist hardware is required. The plexDIA and phosphoproteomics cohorts — processed to demonstrate workflow support rather than benchmarked against deposited data — likewise complete within minutes to hours per cohort (Fig. 2b), even though the deep variable-modification phosphoproteomics search is the most expensive per run. Per-step wall-clock and resource envelopes, and per-run wall-clock broken down by dataset type, are reported in Supplementary Note 4 (Figs. S1–S3). All of the above runs were executed on the EMBL-EBI HPC cluster; to confirm that the workflow is not tied to a single infrastructure, an independent group ran `quantmsdiann` on a separate (non-EBI) SLURM cluster with the same container-pinned profiles, reproducing the per-step parallel and collective runtime structure on third-party infrastructure (Supplementary Fig. S1b).

To confirm that this distributed, SDRF-driven execution matches a single-machine (desktop) `DIA-NN` run, we benchmarked `quantmsdiann` against the ProteoBench community submissions [11], processing the four public ProteoBench DIA modules (Orbitrap Astral, *Module 7*; single-cell Astral, *Module 9*, PXD049412; timsTOF diaPASEF, *Module 5*, PXD062685; SCIEX ZenoTOF 7600, *Module 10*, PXD070049) and comparing quantification accuracy and per-precursor identification counts against the public-submission snapshot (29 May 2026). All runs were library-free (the library predicted in-silico from the FASTA; `quantmsdiann` also supports user-supplied libraries), so for a like-for-like comparison we kept only predicted-from-FASTA `DIA-NN` community submissions, and we report the oldest supported release (1.8.1) and the current 2.5.1-enterprise build at `DIA-NN`'s recommended per-version q-values under a uniform 1% global precursor cut-off. On the two modules with predicted-from-FASTA `DIA-NN` community submissions (Orbitrap Astral, Module 7, $n = 15$; timsTOF diaPASEF, PXD062685, $n = 8$), `quantmsdiann` reproduces the expected HYE per-species $\log_2$ fold-changes, and across both releases its measured ratios sit within the spread of the standalone, desktop `DIA-NN` community submissions (Fig. 2c). We therefore observe no major differences in quantification accuracy between desktop `DIA-NN` and `quantmsdiann`: scaling the analysis across the cluster delivers the same result as a single-machine run.

Identification depth, by contrast, rises monotonically with `DIA-NN` version on every module (precursors +9–22%, per-run protein groups +10–12%, and proteins quantified in every run +11–18% from 1.8.1 to 2.5.1-enterprise; the academic 2.5.1 build and the full per-version, per-module breakdown are in Supplementary Note 5, Fig. S4), in line with the broader ProteoBench community trend [11] and motivating reanalysis of public deposits with the current build (Sections 3.3–3.4).

3.3. Single-cell DIA reanalysis improves with current `DIA-NN` releases

To assess `quantmsdiann` on low-input data, where library-free sensitivity and match-between-runs matter most, we reanalysed two label-free Orbitrap Astral single-cell cohorts, HeLa (PXD046357) and a larger A549/H460 cohort of 146 single cells (PXD049412), through the workflow under `DIA-NN` 1.8.1 and the current 2.5.1-enterprise build, both parametrised from the same SDRF (Fig. 3). The newer 2.5.1-enterprise version increased total identifications: on HeLa Astral, total precursors rose from 22,631 to 26,238 (+16%) and protein groups from 4,137 to 4,840 (+17%), with the gain holding per cell (Fig. 3a). On the deeper A549/H460 cohort the reanalysis reproduced the original study's Astral single-cell depth [19], reaching a median of 3,383 protein groups per cell and up to 5,654 in the deepest cells, with total precursors rising from 61,439 to 72,031 (+17%) and total protein groups from 7,039 to 7,564 (+7%); the protein-level gain is more modest than the precursor gain, consistent with the near-saturated protein depth of this larger cohort. The improvement is largest in the data-completeness profile (the number of protein groups quantified across an increasing fraction of cells; Fig. 3b), consistent with stronger cross-run propagation, while quantitative precision is preserved (Fig. 3c).

3.4. The value of reanalysing public DIA deposits at scale

Public DIA deposits are typically processed once, with the search engine available at the time of deposition, and are seldom revisited — leaving recoverable identifications in raw data that has already been collected, paid for, and published. Reanalysing these deposits with a current `DIA-NN` build recovers them: across every deposit reanalysed here — single-cell, bulk cell-line and spatial — reprocessing yielded more protein groups than the originally deposited or published analysis, and more precursors wherever the original was itself `DIA-NN`, on every deposit (Fig. 4). The recovery is therefore not a property of any single dataset but a systematic return on reprocessing archived DIA data with an up-to-date engine. These comparisons assume that recent `DIA-NN` (2.5/2.5.1) exercises largely correct false-discovery-rate (FDR) control; older releases and other search engines may have substantially higher true FDR at the same nominal 1% q-value, and verifying their FDR calibration on each deposited dataset is beyond the scope of this work. The reported gains should therefore be read as identifications recovered under a current, FDR-controlled engine, not as strictly FDR-matched differences; for the same reason we restrict cross-engine comparisons to identification counts under the original studies' own thresholds and do not compare plexDIA protein groups or phosphoproteomic site numbers across engines (Methods). To show the value of running one SDRF-driven workflow across many independent deposits, we reanalysed five public cell-line DIA panels in a single `quantmsdiann` invocation: NCI-60 (PXD003539, Guo *et al.* 2019 [20]; originally OpenSWATH), ProCan-DepMapSanger (PXD030304, Gonçalves *et al.* 2022 [21]; originally `DIA-NN` 1.7.x), the Sun *et al.* 2023 PCT-SWATH breast-cancer panel (PXD004701, 76 lines [22]), the MultiPro batch-effect testbed (PXD041421, Wang *et al.* 2023 [23]), and Tognetti *et al.* 2021 (PXD017199, 67 breast-cancer lines [24]) (Supplementary Fig. S8). For the two cohorts with a published DIA protein-group headline, the reanalysis recovered more protein groups under the original studies' $\geq$ 2-unique-peptide criterion, applied to the deposited count and to the `quantmsdiann` count under its own global protein-group rule: NCI-60 6,812 versus 4,284 (+59%) and ProCan 8,993 versus 6,698 (+34%; Fig. 4a). This $\geq$ 2-peptide gain exceeds the change in the total protein-group count: current `DIA-NN` releases identify more peptides per protein, so proteins that the original analyses supported with a single peptide now clear the stricter two-peptide threshold, expanding the high-confidence protein set rather than only the marginal one. The Sun PCT-SWATH panel, whose deposited OpenSWATH analysis reports proteotypic proteins rather than a $\geq$ 2-peptide count, shows a comparable recovery at the standard 1% protein-FDR rule: 8,304 `quantmsdiann` protein groups (`Lib.PG.Q.Value` $\leq$ 0.01) versus 6,183 in the deposited analysis (+34%). The size of the gain reflects the age and engine of the deposited analysis, largest for the oldest, OpenSWATH-era NCI-60 (+59%); the recoverable value is therefore greatest exactly where deposits are most numerous: the large back-catalogue of DIA data processed with older or non-`DIA-NN` engines. Each original count is recomputed from public data — NCI-60 and Sun from the deposited PRIDE OpenSWATH analyses, ProCan from the authors' figshare peptide-count matrix — so every bar is reproducible rather than a transcribed paper headline (Methods). (PXD041421 and PXD017199 carry no published DIA headline and contribute reanalysis-only counts.) Across the five cohorts the reanalysis spans 12,914 unique UniProt accessions, of which 5,716 (44.3%) are detected in all five (a pan-cohort core proteome) and a further 1,472 (11.4%) in exactly four. Tissue-level coverage spans 29 tissue categories (27 pan-cancer tissues plus a non-cancer reference and an unassigned bin; Supplementary Fig. S8b), which maps both the cell lines contributing to each tissue and the union of unique proteins detected per tissue, led by lung, breast, and haematopoietic-and-lymphoid tissue. Because every count comes from a single SDRF-parameterised `quantmsdiann` invocation, the resulting protein matrix is queryable from the published QPX archive and can be filtered, joined, or re-grouped by Cellosaurus, NCIt, or Disease Ontology terms without manual harmonisation, turning a set of independently deposited cohorts into a single reusable cross-study resource. The same recovery extends beyond bulk cell lines to another acquisition mode: a spatial Deep Visual Proteomics cohort of MYCN-amplified neuroblastoma recovered more identifications than its original `DIA-NN` 1.8.1 analysis (PXD064049; precursors 21,535 versus 17,958, +20%; protein groups 3,301 versus 2,947, +12%, both counted from the per-precursor reports under the same 1% `Lib` q-value rather than the deposited count matrices, which are written at different run-specific q-values per `DIA-NN` version). Residual protein-group inference inflation is higher in the deposited 1.8.1 analysis (3.2% multi-accession groups) than in the reanalysis (1.3%), so this protein-group gain is a conservative estimate. Per-cohort completeness, peptide-per-protein distributions and accession overlap for the reanalysed cohorts are provided in Supplementary Note 6 (Figs. S5–S8).

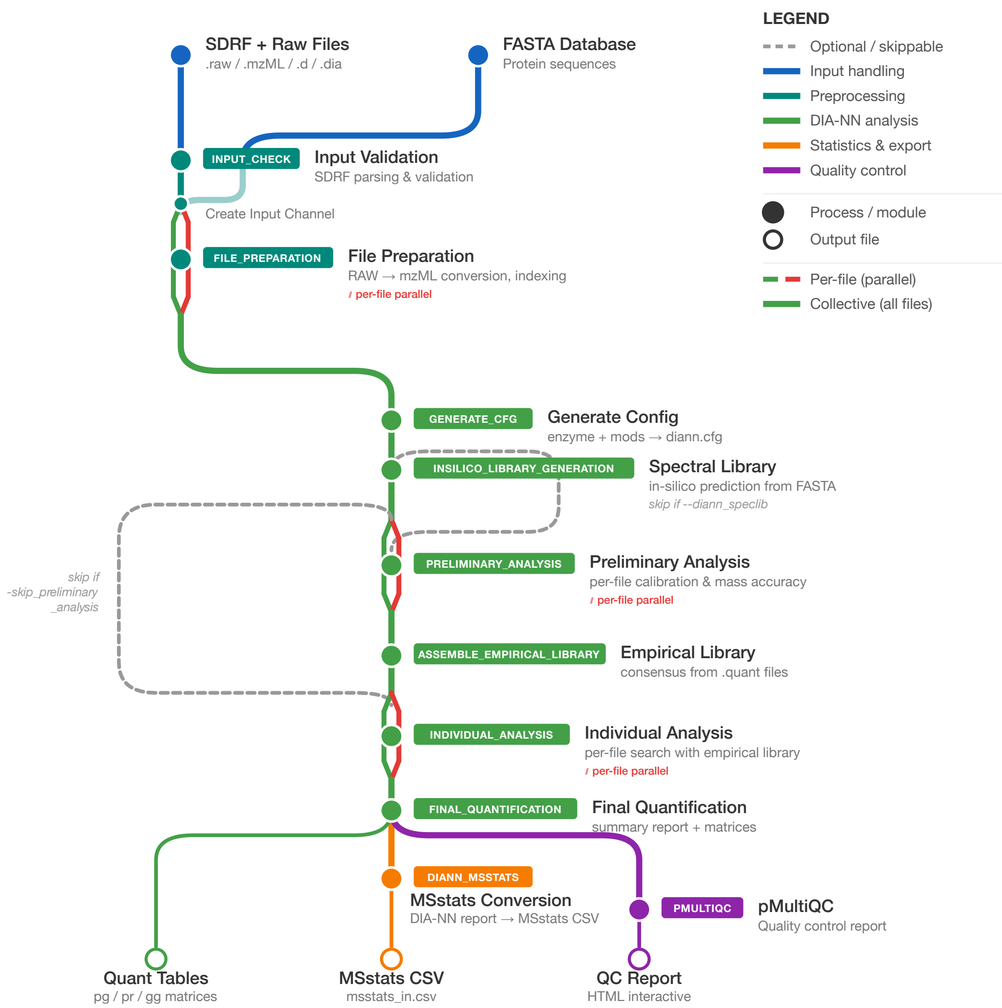

Figure 1: **The `quantmsdiann` workflow.** Subway diagram: mandatory modules as filled circles, optional/skippable modules as open circles. Per-file parallel stages (red; raw-file preparation, preliminary calibration, individual search) run as one task per raw file, whereas collective stages (green; in-silico library generation, empirical library assembly, final quantification, MSstats conversion, `pmultiqc` reporting, and QPX export) run once over the full cohort.

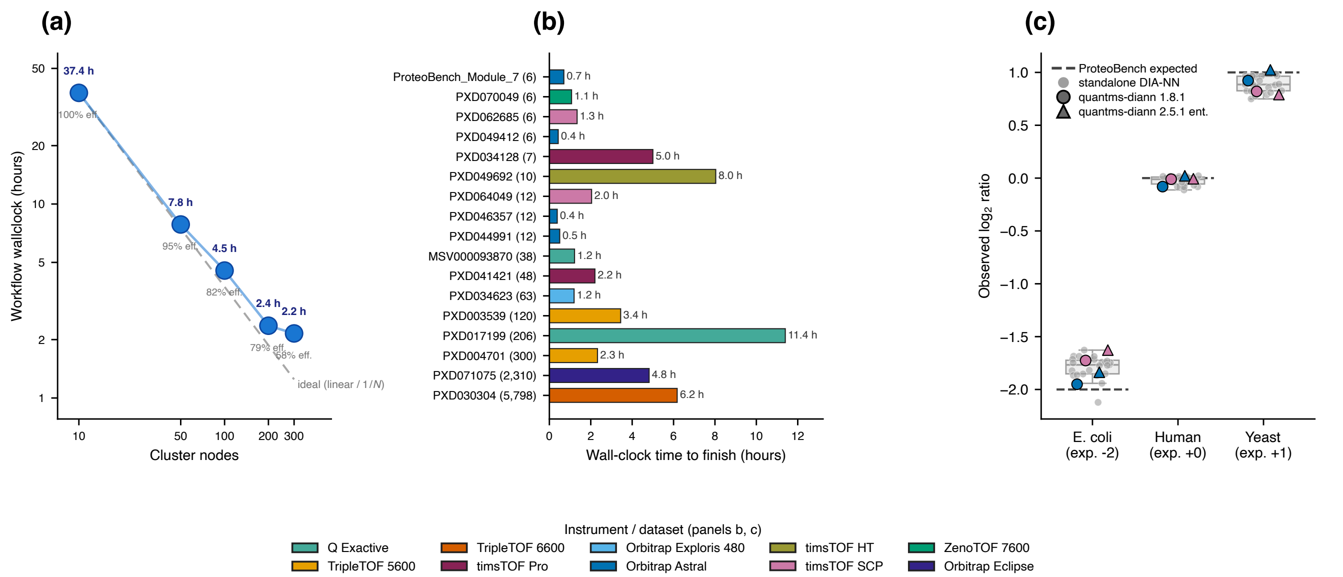

Figure 2: **Runtime scaling and ProteoBench quantification accuracy.** (**a**) Workflow wall-clock versus number of cluster nodes (Nextflow `executor.queueSize`) on the PXD071075 single-cell cohort (log–log; one run per point). The dashed grey line is ideal $1/N$ scaling anchored at 10 nodes; parallel efficiency is printed under each dot. The curve bends above ideal beyond ~100 nodes (efficiency 100% $\rightarrow$ 58%), reflecting fixed serial cohort-level steps and shared-cluster contention on this EMBL-EBI allocation. (**b**) Wall-clock per reanalysis, one bar per dataset, ordered by cohort size and coloured by instrument family. (**c**) ProteoBench accuracy per HYE species: observed $\log_2$ fold-change against the expected ratio (dashed) for the two DIA modules with predicted-from-FASTA `DIA-NN` submissions (Orbitrap Astral, Module 7; timsTOF diaPASEF, Module 5, PXD062685). Standalone community runs (all versions) form the grey strip and box; `quantmsdiann` points are coloured by instrument and shaped by `DIA-NN` version (circle, 1.8.1; triangle, 2.5.1-enterprise), tracking the community distribution, i.e. distributed orchestration does not change quantification accuracy.

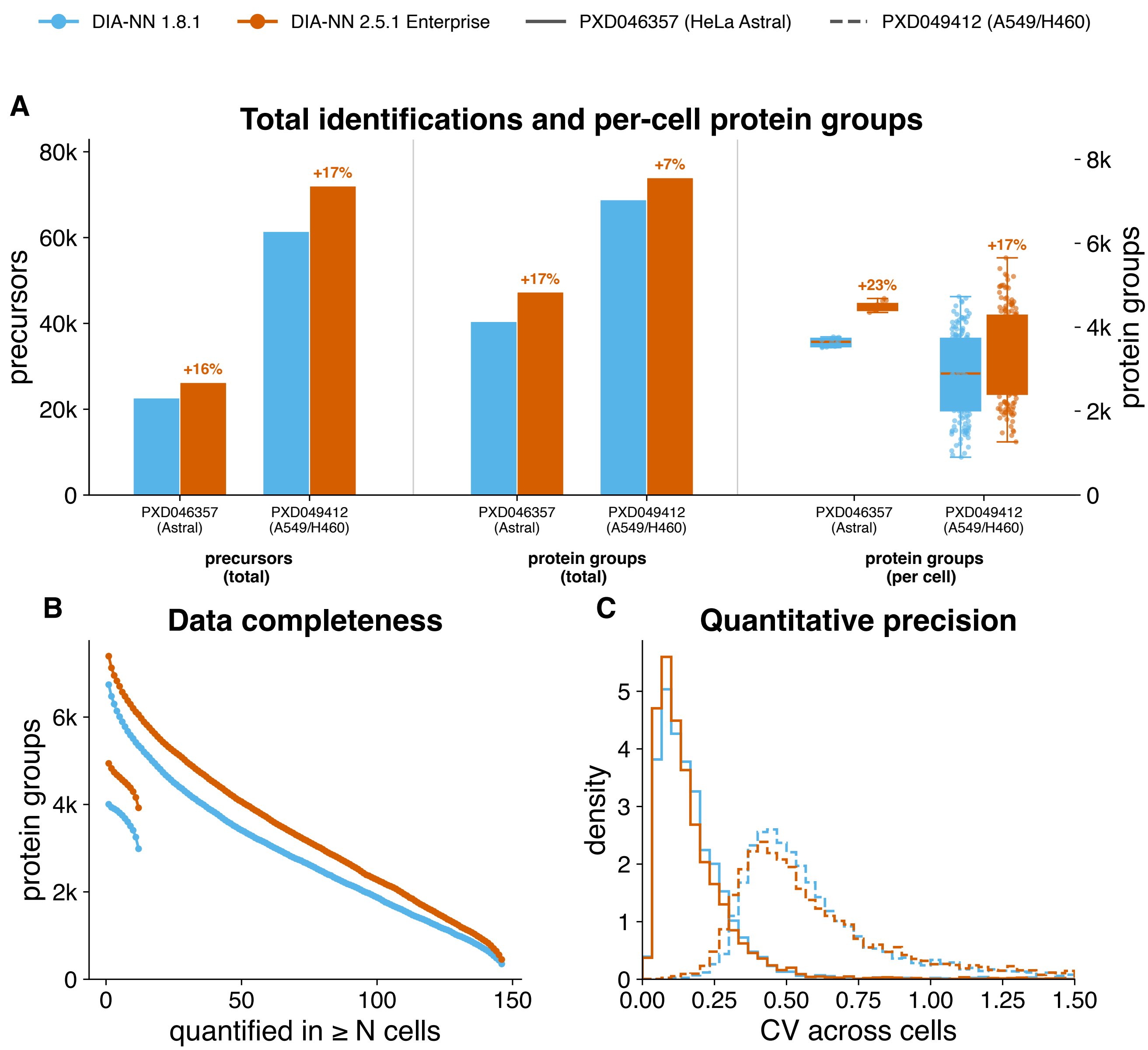

Figure 3: **Single-cell DIA reanalysis by `quantmsdiann`, comparing `DIA-NN` 1.8.1 with the current 2.5.1-enterprise build on identical raw data.** Two label-free Orbitrap Astral cohorts: HeLa (PXD046357) and a larger A549/H460 cohort of 146 single cells (PXD049412). (**a**) Total identifications and per-cell depth per build: total precursors (left axis), total protein groups, and per-cell protein-group distribution (box and jittered points, right axis), with per-build percentage change annotated. A549/H460 reaches comparable per-cell depth on many more cells but is already saturated in total protein groups, so the version effect is concentrated in precursor depth. (**b**) Data-completeness curves (HeLa solid, A549/H460 dashed): protein groups quantified in at least N cells (match-between-runs signature; right endpoint is the complete-profile count). (**c**) Per-protein coefficient of variation across cells (HeLa solid, A549/H460 dashed; axis truncated at CV = 1.5), summarising quantitative precision; the larger A549/H460 cohort shows the expected higher cell-to-cell variability.

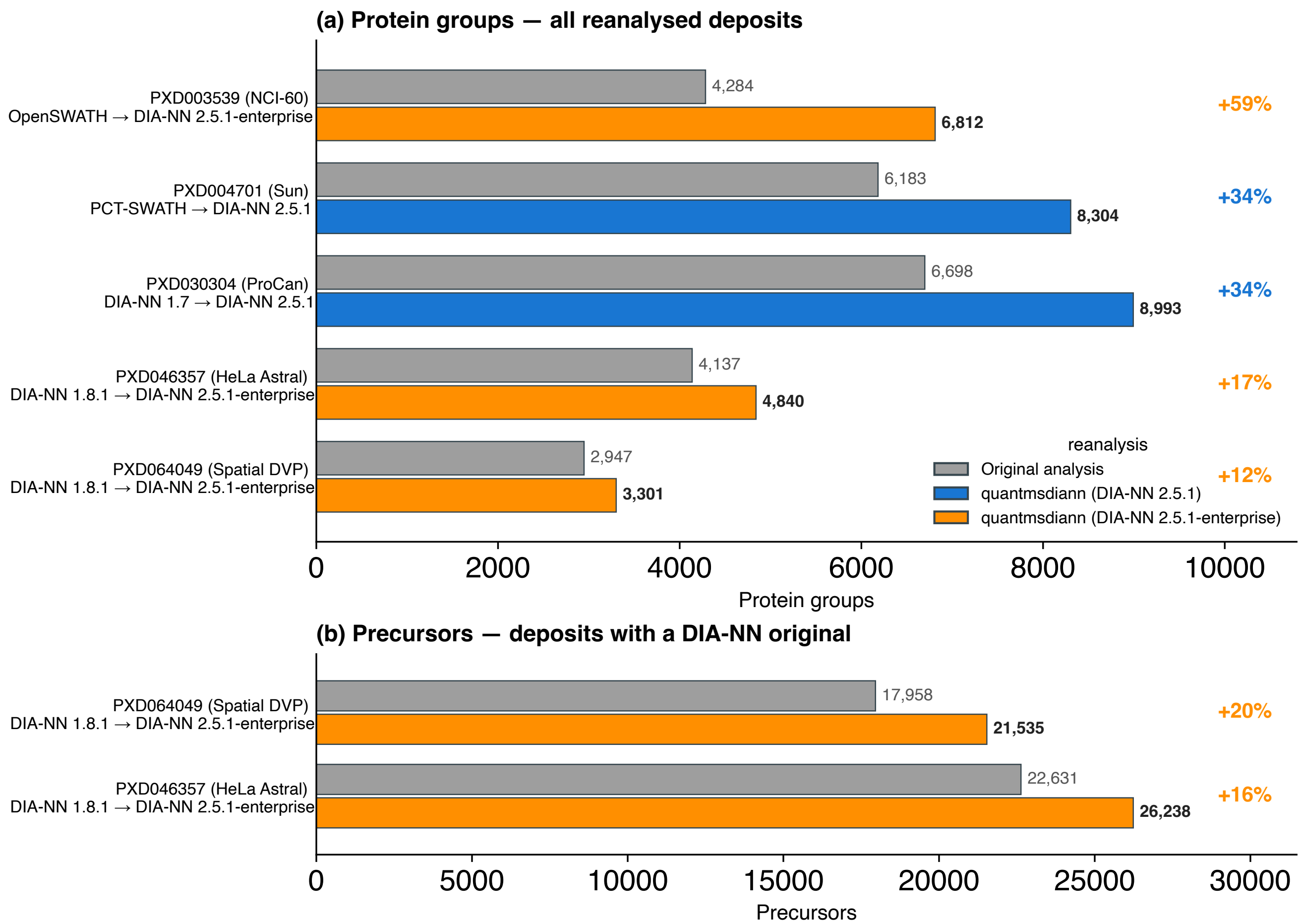

Figure 4: **Reanalysis with current `DIA-NN` recovers more than the originally deposited analyses.** Each public DIA deposit is shown as the originally deposited or published count (grey) versus the `quantmsdiann` reanalysis (coloured by the `DIA-NN` version used: 2.5.1 or 2.5.1-enterprise), sorted by gain. (**a**) Protein groups for the five reanalysed deposits, spanning single-cell (HeLa Astral, PXD046357), bulk cell-line (NCI-60, PXD003539; ProCan, PXD030304; Sun, PXD004701) and spatial Deep Visual Proteomics (PXD064049). The multiplexed plexDIA cohort (MSV000093870) is not shown: plexDIA protein-group counts are not comparable across `DIA-NN` versions. (**b**) Precursors, restricted to the two deposits with a `DIA-NN`-based original precursor count (HeLa Astral and spatial DVP); the OpenSWATH/PCT-SWATH originals provide no comparable precursor count.

4. Discussion

quantmsdiann automates the path from a deposited DIA dataset to a reproducible quantification table, at a scale that supports systematic reanalysis of public DIA archives. The SDRF-first design encourages experiment annotation upstream of analysis and produces metadata that is interoperable with the wider proteomics ecosystem [9]; the QPX archive bundles the SDRF, harmonised quantification tables and pipeline provenance into a single Parquet container, so that analysis results can be distributed, archived, and re-analysed together with their corresponding metadata. By focusing on DIA-NN rather than the multi-engine scope of the related bigbio/quantms pipeline [6], quantmsdiann achieves a leaner dependency surface and faster pipeline-level execution for the DIA-only regime, where the dominant cost is per-file DIA-NN invocation rather than across-engine comparison.

Every reanalysis in this study completed in less than 12 h on the EMBL-EBI HPC cluster, independent of cohort size: the 2,300-run single-cell cohort finished in 2.2 h at 300 nodes. This follows from how end-to-end wall-clock is structured, as a largely fixed serial overhead plus the per-file analysis stages. The serial overhead is dominated by in-silico spectral-library generation from the FASTA (the single largest non-parallel step), plus cohort-level consolidation. The per-file stages run as independent tasks across the cluster (Fig. 2a), so their elapsed time grows only sub-linearly with cohort size as files are processed in parallel waves (per-step envelopes in Supplementary Note 4, Fig. S1). Time-to-finish is therefore dominated by the fixed overhead for small cohorts and stays within a few hours even for the largest deposits, so a small dataset is not necessarily faster than a much larger one (Fig. 2b); the serial overhead itself scales with the predicted-library size, growing for wide variable-modification searches such as phosphoproteomics, rather than with the number of files.

Our evaluation covers three complementary aspects of quantmsdiann: analytical agreement with the desktop version of DIA-NN (ProteoBench; Fig. 2c), recovery against published per-study quantifications (independent reanalyses; Figs. 3, 4), and archive-scale throughput (pan-cohort; Supplementary Fig. S8). On the ProteoBench DIA modules, quantmsdiann's quantification accuracy is comparable to the standalone DIA-NN community submissions: across both DIA-NN releases its per-species HYE fold-changes sit within the spread of the community runs, so distributed, SDRF-driven orchestration reproduces a desktop DIA-NN analysis rather than changing its output. The recovery results show that reanalysis adds the most value for older deposits, those originally processed with earlier or non-DIA-NN engines, while recently processed cohorts gain little because they already approach current performance. An SDRF-driven workflow is well suited to capture this value, because it makes each deposit easy to reprocess as new DIA-NN versions are released. The workflow already supports broader single-cell and laser-microdissected spatial acquisition modes (Fig. 1, library-handling branch), and SDRFs for the Brunner [2], Derks [25], Leduc [26], Wang [4], Mund [27] and Makhmut [28] datasets are in preparation.

The cross-DIA-NN-version reproducibility view enabled by quantmsdiann is, to our knowledge, the first pipeline-level treatment of this question for DIA-NN. We expect this to become increasingly important as DIA-NN evolves: a deposited single-cell DIA dataset analysed with DIA-NN 1.8 today and re-analysed with DIA-NN 2.x in two years should not produce different biological conclusions for the same data and parameters. In current practice such re-analyses are rarely performed, because the bespoke per-study scripts make them prohibitively expensive, a barrier the single-invocation audit of Section 3.2 removes. By design, quantmsdiann ends at harmonised quantification, leaving batch correction, cell-type assignment, and differential-abundance analysis to downstream packages such as scp [29] that read its QPX or MSstats output directly; in this sense it complements existing single-cell DIA workflows rather than competing with them. `quantmsdiann` is open-source under the MIT license and is openly developed as a community effort within the nf-core and bigbio proteomics ecosystem, so that it can continue to track new `DIA-NN` releases and acquisition modes as they appear.

Data Availability

All datasets reanalysed in this study are publicly available through ProteoMeXchange. Cell-line bulk DIA reanalyses: PRIDE PXD003539 (Guo 2019, NCI-60), PXD004701 (Sun 2023), PXD030304 (ProCan-DepMapSanger), PXD017199 (Tognetti 2021), PXD041421 (Wang 2023). Spatial Deep Visual Proteomics reanalysis: PRIDE PXD064049 (MYCN-amplified neuroblastoma, diaPASEF). Scalability experiment: PRIDE PXD071075 (Wu 2025). ProteoBench benchmarks: PXD049412 (single-cell, Module 9), PXD062685 (diaPASEF, Module 5), PXD070049 (ZenoTOF, Module 10), and Module 7 (Astral) datasets. The full list of PRIDE accessions used in the pan-cohort of DIA cell-line reanalyses is provided in Supplementary Note 2 (Table S1). `quantmsdiann` is available at `https://github.com/bigbio/quantmsdiann` under the MIT license; documentation is hosted at `https://quantmsdiann.quantms.org/`. The pipeline is implemented in Nextflow DSL2 ($\geq$25.10.4) and packages each step as a versioned Docker, Singularity, or Apptainer container image. `DIA-NN` 1.8.1 is distributed publicly via BioContainers; because the `DIA-NN` license restricts redistribution of later releases, containers for `DIA-NN` 1.9 and above are built locally from the recipes in `https://github.com/bigbio/quantms-containers` (see Experimental Procedures). Pipeline version v2.2.0 (release codename "Chongqing", 2026-06-16) was used for the analyses reported in this paper. Analysis scripts that generated the figures in this manuscript are available at `https://github.com/ypriverol/quantmsdiann-paper` (this repository).

Supplemental Data

This article contains supplemental data. The Supplementary Notes provide the `DIA-NN` container-build recipes (Supplementary Note 1), the full PRIDE accession list (Supplementary Note 2, Table S1), the filtering and protein-counting rules (Supplementary Note 3), per-process runtime and resource envelopes (Supplementary Note 4, Figs. S1–S3), the ProteoBench per-version identification breakdown (Supplementary Note 5, Fig. S4), per-cohort completeness, peptide-per-protein and accession-overlap figures (Supplementary Note 6, Figs. S5–S8), and a table of software, formats, and public resources with their repositories (Supplementary Note 7, Table S2).

Conflict of Interest

V.D. holds shares of Aptila Biotech. The other authors declare no competing interests.

Acknowledgments

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Author Contributions

QXY, YS and CD developed the pipeline. HW helped build the containers and contributed to the workflow implementation and manuscript writing. AL annotated the SDRF metadata and helped run the workflow. TS, FC, NS and JN assisted with the DIA-NN reanalyses and manuscript writing; TS additionally contributed to the workflow design and to the design of QPX and the SDRF format, and JN tested the tool and identified multiple issues. OK provided feedback on the plexDIA datasets. VD provided DIA-NN expertise and guidance on cross-version reproducibility. JAV provided proteomics data-resource infrastructure and funding, supervised the EMBL-EBI contributors, and contributed to manuscript review and editing. MB contributed to manuscript writing and supervised the students; MB and YPR conceived and supervised the project. QXY and YPR wrote the original draft, with input from all authors. All authors read and approved the final manuscript.

Abbreviations

CLI command-line interface
DDA data-dependent acquisition
DIA data-independent acquisition
DIA-NN data-independent acquisition neural networks (DIA search and quantification software)
diaPASEF data-independent acquisition with parallel accumulation–serial fragmentation
EFO Experimental Factor Ontology
FDR false discovery rate
HPC high-performance computing
HUPO-PSI Human Proteome Organization Proteomics Standards Initiative
HYE human, yeast, and *E. coli* (ProteoBench three-proteome mixture)
IQR interquartile range
LSF Load Sharing Facility (HPC scheduler)
MS mass spectrometry
MS1/MS2 first-/second-stage mass spectrometry (precursor/fragment scans)
PASEF parallel accumulation–serial fragmentation
PSI-MS HUPO-PSI mass spectrometry controlled vocabulary
QPX Quantitative Proteomics eXchange
SDRF Sample and Data Relationship Format

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